

Mohamed Abdellatif Etman

Cairo, Egypt · +201024963291 · mae25806@gmail.com · linkedin.com/in/mohamed-etman-7a307b218
· github.com/mohamedabdalltif

PROFESSIONAL SUMMARY

Innovative AI Engineer with expertise in designing and deploying end-to-end solutions using Generative AI, multimodal analysis, and cloud technologies. Proven ability to build and scale complex AI systems, leveraging state-of-the-art models (GPT-4o, Gemini, Llama 3), RAG pipelines, and vector databases to solve real-world business problems. Adept at full-stack development and MLOps, with a strong focus on creating robust, scalable, and efficient applications on Microsoft Azure.

TECHNICAL SKILLS

AI & Machine Learning:

- **Generative AI:** Large Language Models (GPT-4o, Gemini, Llama 3, Qwen-VL), Retrieval-Augmented Generation (RAG), Vector Databases (FAISS), LangChain.
- **Deep Learning & NLP:** Transformers, BERT, CNNs, RNNs, PyTorch, TensorFlow, Keras, spaCy, NLTK.
- **Computer Vision & Audio:** OpenCV, OpenPose, MediaPipe, Whisper, Praat, librosa.

Programming & Development:

- **Languages:** Python, JavaScript, TypeScript, PHP, SQL.
- **Backend:** FastAPI, Django, Flask, Redis.
- **Databases:** MongoDB, MySQL.
- **Frontend & GUI:** React Native, Streamlit, PyQt5.

Cloud & DevOps:

- **Cloud Platforms:** Microsoft Azure (Cognitive Services, App Service).
- **Tools:** Docker, Git, GitHub, REST APIs, Bash Scripting.

WORK EXPERIENCE

AI Engineer (Freelance)

Remote

Dec 2024 - Present

- Designed and delivered end-to-end AI applications powered by Generative AI models (e.g., GPT-4o, Gemini), integrating Retrieval-Augmented Generation (RAG) pipelines and vector databases for contextual understanding.
- Built scalable NLP services supporting multiple languages using Azure Cognitive Services (Text Analytics, Translator, Speech) and custom transformer pipelines.
- Developed backend APIs using FastAPI, with session handling and automation workflows via Redis and RESTful service integration.
- Created interactive dashboards and monitoring tools using Streamlit, and deployed complete solutions to Microsoft Azure cloud infrastructure.

AI Developer – MCIT (Gov. Innovation Project)

Remote, Egypt

Ministry of Communications and Information Technology (MCIT)

Nov 2024 - Mar 2025

- Developed a multimodal AI system for employee performance evaluation, integrating video, audio, and text analytics with Generative AI models (Llama 3, Qwen-VL) and Vector Databases.
- Utilized OpenPose and MediaPipe for gesture recognition, and Whisper for audio processing, deploying solutions on Azure.
- Compared open-source AI pipelines with Gemini 2.0 Flash to optimize efficiency, achieving startup prototype recognition.

GenAI Engineer (Internship)

Remote, Egypt

National Bank of Egypt (NBE)

July 2024 - Oct 2024

- Built a Financial Dashboard Auto Transcript Analyzer using Gen-AI to extract data from chart images, generate summaries, and answer user queries, simulating a RAG pipeline.
- Integrated Transformers (Donut, VisionEncoderDecoder) for data extraction and GPT-NeoX-20B for narrative generation.

- Led the development of mobile applications using React Native and PHP, focusing on scalable architecture and robust backend services.
- Managed database schemas and optimized queries for performance in data-intensive applications.

PROJECTS

AI-driven Video Processing Agent (Ministry of Communications and Information Technology)

- **Overview:** Developed a multimodal AI system to analyze video, audio, and text for generating structured employee performance reports. The project’s impact led to its recognition as a startup concept.
- **Features:** Implemented video analysis (OpenPose, MediaPipe), audio processing (Whisper, Praat), and text analytics (spaCy). Leveraged LLMs (Llama 3, Qwen-VL) for automated report generation.
- **Technologies:** Python, PyTorch, OpenPose, MediaPipe, Whisper, spaCy, Llama 3, Gemini 2.0 Flash, Azure.

Smart WhatsApp Assistant (Freelance for Saudi Retail Company)

- **Overview:** Architected and deployed an end-to-end conversational AI assistant using a Retrieval-Augmented Generation (RAG) pipeline to handle customer interactions in Arabic and English.
- **Features:** Utilized Generative AI (Gemini, GPT-4o) and a FAISS vector database for context-aware responses. Integrated Twilio API, FastAPI, and Redis for messaging and session management. Deployed on Microsoft Azure.
- **Technologies:** Gemini, GPT-4o, RAG, FAISS, LangChain, Azure Cognitive Services, FastAPI, Redis, Streamlit, Twilio API.

Financial Dashboard Auto Transcript Analyzer (National Bank of Egypt)

- **Overview:** Built a Gen-AI tool to analyze dashboards by extracting data from chart images and generating summaries. The system simulates a RAG pipeline to provide intelligent answers to user queries.
- **Features:** Used Vision-Encoder-Decoder models for chart data extraction and a fine-tuned QA model for query answering. Deployed as a web API with FastAPI and included a PyQt5 desktop app for screen capture.
- **Technologies:** Transformers, RAG-style Retrieval, GPT-NeoX-20B, FastAPI, Azure Cognitive Services, PyQt5, Torch.

EDUCATION

Diploma in AI, Data Science, and Cloud Computing

Digital Egypt Builders Initiative (DEBI), Ministry of Communication

Hybrid, Egypt

Jan 2024 - Dec 2024

Bachelor’s Degree in Computer Science

Faculty of Computers and Information

Tanta, EGYPT

July 2019 - July 2023

CERTIFICATIONS

- **Azure:** AI-102: AI Engineer Associate.
- **DEBI Certifications:** Stanford Machine Learning Specialization, Deep Learning Specialization, Generative AI with Large Language Models, MLOps Tools (MLflow, Hugging Face).

LANGUAGES

- Arabic: Native
- English: Professional Working Proficiency

PERSONAL INFORMATION

- Military Status: Exempted